

WT - Exp - 3

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Project name: Water Turbidity Detector

Aim:

Develop an automated system that accurately measures and detects the turbidity level of water to ensure water quality monitoring and safety.

Introduction:

Water turbidity, which refers to the cloudiness or haziness of a liquid due to suspended particles, is a critical parameter in water quality assessment. High turbidity can indicate the presence of contaminants, microorganisms, or pollutants, making it necessary to monitor water conditions in real time. Manual testing methods are often time-consuming and prone to errors, making automated detection a more efficient approach.

This project aims to develop a Water Turbidity Detector using sensors and an Arduino-based system. The system will employ a turbidity sensor to measure water clarity and an Arduino microcontroller to process the data and display results. The detection mechanism can be used in water treatment plants, environmental monitoring, and industrial applications to ensure safe and clean water supply.

Objectives:

- **Automate Water Quality Assessment:** Design a system that efficiently measures water turbidity without manual intervention.
- **Enhance Accuracy and Efficiency:** Implement real-time sensor-based data collection for precise turbidity measurement.
- **Provide a Cost-Effective and Scalable Solution:** Develop a system that is affordable and can be adapted for different water monitoring applications.

Challenges:

- **Sensor Calibration and Accuracy:** Variability in sensor readings due to different types of suspended particles.
- **Environmental Conditions:** Temperature, light interference, and water flow may impact sensor performance.
- **Real-Time Data Processing:** Ensuring fast and accurate analysis of sensor readings.

Solution Approach:

1. **Object Detection:** The ultrasonic sensor detects when an object is placed on the system.
2. **Moisture Measurement:** The soil moisture sensor determines the object's moisture content.
3. **Data Processing:** The Arduino processes the sensor readings and decides the category.
4. **Object Segregation:** A motorized mechanism, positioned behind the object, rotates left for wet objects and right for dry objects, directing them accordingly.

Scope:

- **Drinking Water Quality Monitoring:** Ensuring safe drinking water by detecting contamination.
- **Industrial Wastewater Management:** Monitoring effluent discharge to comply with environmental regulations.
- **Aquatic Ecosystem Monitoring:** Assessing water conditions in lakes, rivers, and reservoirs.

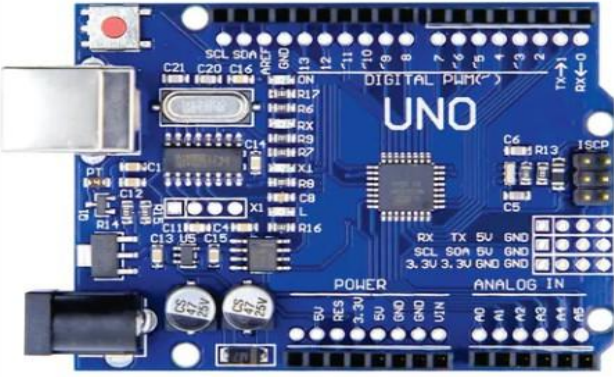
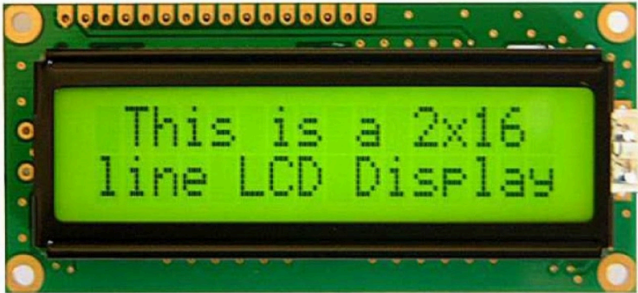
Advantages:

- **Automated and Real-Time Monitoring:** Reduces manual effort and provides instant results.
- **High Accuracy:** Uses sensor-based measurements for precise turbidity analysis.
- **Compact and Cost-Effective:** Simple design with minimal hardware requirements.
- **Scalability:** Can be integrated into larger water quality monitoring systems.

Conclusion:

This Arduino-based Water Turbidity Detector provides an efficient, real-time solution for monitoring water quality. By integrating a turbidity sensor with an automated processing and display system, the project ensures precise and reliable detection of water clarity levels. The system is scalable, cost-effective, and suitable for various water monitoring applications, contributing to better environmental and public health management.

Components:

Sr no.	Description	Image
1.	Arduino Uno SMD A microcontroller board based on the ATmega328P, designed for embedded applications with digital and analog I/O pins. It processes data from sensors and controls the servo motor to classify objects as wet or dry.	 The image shows an Arduino Uno SMD microcontroller board. It is a blue printed circuit board (PCB) with various components including a USB Type-B port, a DC power jack, a reset button, and a large ATmega328P microcontroller. The board is populated with surface-mount components and has a standard pin header at the bottom.
2.	Turbidity Sensor A sensor that detects water clarity by measuring the light transmission and scattering properties of suspended Particles	 The image shows a turbidity sensor module. It consists of a blue plastic housing with two clear lenses for light transmission and scattering measurement. The module is connected to a black PCB with various electronic components, including a blue potentiometer and several resistors. It has a standard 5-pin header for connection to a microcontroller.
3.	LCD Screen A display unit that shows turbidity level in real time helping users access water quality.	 The image shows a 2x16 line LCD display. It is a green printed circuit board (PCB) with a black LCD screen. The screen displays the text "This is a 2x16 line LCD Display" in green characters. The PCB has a standard 16-pin header on the right side.

4.	USB Cable A standard cable used to supply power and enable data communication between the Arduino and a computer. It facilitates programming and real-time monitoring of the system.	
5.	M-F Jumpers Flexible connectors with male-to-female terminals that establish electrical connections between the Arduino, sensors, and motor. They ensure proper signal transmission and circuit functionality.	